LegalEase

Phase-Wise Project Document

Project: Template-Based Legal Document Generator with AI-Ready Architecture

Project Type: Web Technology / AI Application

Technology: Python, Flask, ReportLab, HTML5, CSS3

Academic Year: 2026–2027

Project Team: Pavithra D, Sandhiya S, Swathi S, Janani R, Vishnu Priya

Phase 1 – Brainstorming & Ideation

1. Objective/Purpose

Identify the problem of preparing consistently formatted first drafts of common legal agreements and develop the LegalEase concept.

2. Activities/Tasks

• Identified repetitive typing, manual formatting, clause organisation and inconsistency problems.

• Studied manual/word-processor templates and online form-based document services.

• Defined LegalEase as a guided form-based document generator.

• Identified Rental, NDA, Employment and Leave-and-License agreements.

• Considered an optional AI-based clause-refinement layer.

3. Deliverables/Outputs

• Problem statement, project concept, objectives and scope.

4. Timeline/Duration

• Academic Year 2026–2027. Specific phase duration: Not specified.

5. Responsible Person/Team

• Project team: Pavithra D, Sandhiya S, Swathi S, Janani R, Vishnu Priya. Individual responsibilities: Not specified.

6. Dependencies

• Identification of the document-generation problem and existing/manual approaches.

7. Milestones/Completion Criteria

• Problem statement, objectives and scope documented; proposed system concept defined.

8. Status/Progress

• Completed/documented in the project report.

Phase 2 – Requirement Analysis

1. Objective/Purpose

Define the functional and technical requirements of the LegalEase prototype.

2. Activities/Tasks

• Identified document type, name, other party, purpose, duration, amount, deposit, notice and place as inputs.

• Defined validation and sanitisation requirements.

• Defined hardware/software requirements: laptop/desktop, 4 GB RAM or more, modern processor, Windows/Linux/macOS, Python 3.9+, Flask 3.x, ReportLab 4.x, HTML5/CSS3 and modern browser.

• Identified optional Anthropic package/API key for the AI layer.

• Defined in-scope and out-of-scope features.

3. Deliverables/Outputs

• System requirements, functional input requirements, validation requirements, scope and error-handling requirements.

4. Timeline/Duration

• Specific duration: Not specified.

5. Responsible Person/Team

• Project team: Pavithra D, Sandhiya S, Swathi S, Janani R, Vishnu Priya. Individual responsibilities: Not specified.

6. Dependencies

• Python, Flask, ReportLab environment and modern browser; internet/API access only for optional AI layer.

7. Milestones/Completion Criteria

• System requirements, required inputs, scope and validation/error requirements documented.

8. Status/Progress

• Completed/documented.

Phase 3 – Project Design

1. Objective/Purpose

Design the architecture and internal structure of the LegalEase application.

2. Activities/Tasks

• Designed Browser → Flask routes → validators.py → clauses.py → Preview → pdf_utils.py architecture.

• Defined modules for UI, routing, validation, draft generation, preview, PDF, optional AI and testing.

• Created Level 0 and Level 1 Data Flow Diagrams.

• Designed a proposed expanded data model.

• Defined the document-generation algorithm.

• Designed AI integration with template fallback and human review.

• Planned XML escaping before ReportLab processing.

3. Deliverables/Outputs

• System architecture, module structure, DFDs, data model, algorithm and project folder structure.

4. Timeline/Duration

• Specific duration: Not specified.

5. Responsible Person/Team

• Project team. Individual design responsibilities: Not specified.

6. Dependencies

• Requirements from Phase 2; Flask, Python modules and ReportLab.

7. Milestones/Completion Criteria

• Architecture, modules, DFDs, algorithm and folder structure defined.

8. Status/Progress

• Completed/documented.

Phase 4 – Project Planning

1. Objective/Purpose

Organise the project into manageable modules and establish the implementation and testing structure.

2. Activities/Tasks

• Planned app.py, validators.py, clauses.py, pdf_utils.py, ai_layer.py, HTML templates and tests/test_app.py.

• Planned deterministic templates as the primary document-generation method.

• Planned the optional AI layer with template fallback.

• Planned automated testing using Python unittest.

• Planned local Flask execution.

3. Deliverables/Outputs

• Project folder structure, module organisation, testing structure and development/run commands.

4. Timeline/Duration

• Specific planning duration: Not specified; academic year 2026–2027.

5. Responsible Person/Team

• Project team. Specific task allocation: Not specified.

6. Dependencies

• System requirements and project architecture; Python/Flask/ReportLab environment.

7. Milestones/Completion Criteria

• Project structure, modules, testing framework and execution method defined.

8. Status/Progress

• Completed/documented.

Phase 5 – Project Development

1. Objective/Purpose

Implement the LegalEase web application and document-generation functionality.

2. Activities/Tasks

• Developed Flask routes /, /generate and /download.

• Implemented input validation, sanitisation, defaults and amount validation.

• Implemented four agreement templates.

• Developed preview and PDF generation using ReportLab.

• Added signature/witness blocks and educational disclaimer.

• Implemented XML escaping for safe PDF generation.

• Developed optional AI layer with template fallback.

• Created HTML input and result templates.

3. Deliverables/Outputs

• Functional LegalEase prototype, four document templates, input form, preview page, downloadable PDF and optional AI refinement module.

4. Timeline/Duration

• Specific development duration: Not specified.

5. Responsible Person/Team

• Project team. Individual coding responsibilities: Not specified.

6. Dependencies

• Python 3.9+, Flask 3.x, ReportLab 4.x, HTML5/CSS3 and optional Anthropic package/API key.

7. Milestones/Completion Criteria

• Complete source code, four agreement types, preview, PDF generation and optional AI layer implemented.

8. Status/Progress

• Implemented; core version works without an AI API.

Phase 6 – Project Testing

1. Objective/Purpose

Verify that the application handles valid/invalid inputs correctly and generates PDFs successfully.

2. Activities/Tasks

• Created 10 automated unit tests using Python unittest.

• Tested home page, valid rental preview, defaults, invalid/negative amounts, same-party names, unknown type, all four PDF types, special characters and long-text truncation.

• Recommended manual checks of the downloaded PDF, mobile-browser layout and repeated testing after template/AI-prompt changes.

3. Deliverables/Outputs

• tests/test_app.py and test results; 10 automated tests passed.

4. Timeline/Duration

• Specific testing duration: Not specified.

5. Responsible Person/Team

• Project team. Specific testing responsibilities: Not specified.

6. Dependencies

• Completed application modules, Python unittest, templates and AI prompts where applicable.

7. Milestones/Completion Criteria

• 10/10 automated tests passed.

8. Status/Progress

• Completed – 10/10 automated tests passed; manual tests recommended before viva.

Phase 7 – Project Documentation

1. Objective/Purpose

Record the project's concept, requirements, design, implementation, testing, results, limitations and future enhancements.

2. Activities/Tasks

• Prepared the project report.

• Documented abstract, introduction, problem statement, objectives, scope, existing/proposed systems, requirements, analysis, architecture, modules, DFDs, data model, algorithm, source code, testing, output screens, advantages/limitations, future enhancements, viva Q&A;, conclusion and bibliography.

• Documented the educational/legal disclaimer and output-screen descriptions.

3. Deliverables/Outputs

• Complete project report, source-code documentation, testing results, output-screen documentation, viva Q&A; and bibliography.

4. Timeline/Duration

• Specific documentation duration: Not specified; academic year 2026–2027.

5. Responsible Person/Team

• Project team. Specific documentation responsibilities: Not specified.

6. Dependencies

• Information from previous phases, source code, testing results and application outputs.

7. Milestones/Completion Criteria

• Project report, testing results, viva Q&A; and bibliography completed.

8. Status/Progress

• Completed/documented.

Phase 8 – Project Demonstration

1. Objective/Purpose

Demonstrate the complete flow from user input to generated preview and downloadable PDF.

2. Activities/Tasks

• Demonstrate the input screen and document-type selection.

• Enter party and document details and generate the draft.

• Display numbered clauses and download the PDF.

• Show PDF title, clauses, signature block, witness lines, generation date and disclaimer.

• Prepare viva explanations covering AI, deterministic vs generative output, hallucination prevention, validation, XML escaping, Flask, PDF generation and legal status.

3. Deliverables/Outputs

• Generated draft preview, downloadable PDF, demonstration screens and viva questions/answers.

4. Timeline/Duration

• Specific demonstration duration: Not specified; manual tests recommended before viva.

5. Responsible Person/Team

• Project team. Specific presenter allocation: Not specified.

6. Dependencies

• Working application, completed templates, successful testing and PDF generation.

7. Milestones/Completion Criteria

• Input-to-PDF flow documented; viva preparation completed; manual testing recommended before viva.

8. Status/Progress

• Demonstration material documented; report notes that running-application screenshots should be inserted.

Summary Table

Phase	Main Activities	Deliverables	Milestones
1. Brainstorming & Ideation	Problem identification, concept, objectives, scope	Problem statement, concept, objectives, scope	Concept defined
2. Requirement Analysis	Inputs, validation, hardware/software, scope	Requirements and validation specifications	Requirements documented
3. Project Design	Architecture, modules, DFDs, data model, algorithm	Architecture, DFDs, modules, algorithm	System design documented
4. Project Planning	Module structure, folder structure, testing plan	Project structure and testing structure	Development structure established
5. Project Development	Flask, validation, templates, preview, PDF, AI layer	Working prototype and PDF generator	Core application implemented
6. Project Testing	Automated tests and recommended manual tests	Test file and results	10/10 automated tests passed
7. Project Documentation	Report, source code, results, limitations, future work	Complete project report	Documentation completed
8. Project Demonstration	Input → preview → PDF and viva preparation	Preview, PDF, demo material, viva Q&A;	Demonstration flow documented

Overall Project Status: The report describes LegalEase as a completed and tested academic prototype. The core version uses deterministic templates and works without an AI API; the optional AI layer is designed for controlled future clause refinement. Generated documents are educational drafts and require professional legal review before real-world use.